

p -adic Variational Quantum Circuits Have No Barren Plateaus

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Abstract

Barren plateaus, the exponential concentration of gradients in random parameterised quantum circuits, are the central trainability obstruction in variational quantum machine learning. We show that the phenomenon does not survive the passage from complex to p -adic scalars.

The barren-plateau theorem rests on three ingredients: a compact ensemble carrying a Haar probability measure, moment data supplied by unitary t -designs, and concentration in the Archimedean absolute value. Over a quadratic extension K of $\mathbb{Q}_p$ each of the three breaks, and they break in different directions. The first fails outright: the unitary group of K^N is non-compact for $N \geq 2$, so it carries no Haar probability measure and “Haar-random circuit” has no meaning; we exhibit an explicit unbounded family at $p = 3$. The compact replacement is the group $U_N(\mathcal{O})$ of integral unitaries, which contains every circuit realisable from the known p -adic gate sets. The second ingredient is not missing but redundant: Haar averages over $U_N(\mathcal{O})$ push forward, level by level, to counting measures on the finite unitary groups $U_N(\mathcal{O}/\mathfrak{m}^k)$, and a character sum over $U_N(\mathbb{F}_{p^2})$ supplies in closed form the moment data that the complex theory obtains from t -designs. The third ingredient reverses. For an n -qubit circuit whose observable–generator commutator splits over $\mathbb{F}_p$, the gradient attains the largest absolute value the p -adic convergence radius allows, with probability at least $1 - p^{-1} - 10p^{1-r}$, where r measures how far the observable and the generator are from commuting modulo p . The bound does not depend on n , and r grows like 2^n , so the correction is doubly exponentially small exactly where the complex plateau is most severe. We give a standard observable–generator pair, Pauli X against a phase rotation, that achieves $r = 2^{n-1}$.

The splitting hypothesis is a real restriction, not a formality: we exhibit a Hermitian matrix over $\mathbb{F}_9$ that violates it. Subject to it, trainability in the p -adic model is limited by the conditioning of the optimiser, not by gradient concentration.

Keywords

quantum machine learning, barren plateaus, variational quantum circuits, p -adic numbers, non-Archimedean analysis, trainability

1. Introduction

A variational quantum circuit prepares $U(\theta)\psi_0$ and tunes θ to minimise $\mathcal{L}(\theta) = \langle U(\theta)\psi_0, MU(\theta)\psi_0 \rangle$ for an observable M . McClean et al. [1] showed that if $U(\theta)$ is drawn from an ensemble forming a unitary 2-design on $U(2^n)$, then $\mathbb{E}[\partial_\theta \mathcal{L}] = 0$ and $\text{Var}[\partial_\theta \mathcal{L}] = O(2^{-n})$. Gradients therefore concentrate exponentially: the landscape is flat almost everywhere, and the number of measurements needed to resolve a descent direction grows exponentially in the qubit count. Later work traced the effect to the cost function [2] and to ansatz expressibility [3], and it remains the principal argument against scalable variational quantum machine learning.

Read carefully, the theorem is a statement about the real numbers. It says that a real quantity, the gradient, is small in the Archimedean absolute value, with high probability under the Haar measure of a compact group whose volume concentrates as the dimension grows. Three separate facts are doing the work: the ensemble is a compact group and so carries a Haar probability measure; the second moment of that measure is computable because deep circuits approximate 2-designs; and the resulting variance is small in an absolute value for which “small” is a limiting statement. None of the three is a fact about quantum mechanics. All three are facts about $\mathbb{C}$.

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A parallel line of work replaces those scalars. Following Aniello, Mancini and Parisi [4, 5, 6], one rebuilds quantum kinematics over a quadratic extension $K = \mathbb{Q}_{p,\mu}$ of the field $\mathbb{Q}_p$ of p -adic numbers. States are vectors in K^N , gates are unitaries for a K -valued Hermitian form, and the geometry is *ultrametric*: $|x + y|_p \leq \max\{|x|_p, |y|_p\}$. The motivation is that ultrametric geometry is the intrinsic geometry of hierarchical data – taxonomies, phylogenies, file systems – which classical p -adic learning theory has exploited for decades [7, 8, 9, 10]. The quantum side of the programme now has a qubit [11], a tensor product [12], a universal gate set [13], and a first quantum neural network [14].

Whether barren plateaus survive the substitution appears to be open, and the question needs care before it can even be asked. The complex statement concerns a variance, and a variance decaying like 2^{-n} has no p -adic meaning: the p -adic absolute value takes only the discrete values $p^{\mathbb{Z}}$, and a quantity is small when it is divisible by a high power of p (Section 2). The faithful translation asks how often the gradient attains the largest absolute value it can have. We state the question in the form in which this paper answers it, with every term defined in the statement.

Question 1.1 (Barren plateaus over K). *Let p be an odd prime, let K be the unramified quadratic extension of $\mathbb{Q}_p$, with ring of integers $\mathcal{O}$ and non-trivial automorphism $z \mapsto \bar{z}$, and let $N = 2^n$. Write U^* for the conjugate transpose, $\langle v, w \rangle = \sum_i \bar{v}_i w_i$ for the standard Hermitian form on K^N , and let G_N be the group of matrices U with entries in $\mathcal{O}$ and $U^*U = \mathbf{1}$, with Haar probability measure ν . Fix an observable $M = M^*$ and a generator $A = -A^*$ with entries in $\mathcal{O}$, a vector $\psi_0 \in \mathcal{O}^N$ with $\langle \psi_0, \psi_0 \rangle = 1$, and a block $U_B \in G_N$. For $U_A \in G_N$ and $\theta \in \mathbb{Z}_p$ consider the circuit and the loss*

$$U(\theta) = U_B \exp_p(p\theta A) U_A, \quad \mathcal{L}(\theta) = \langle U(\theta)\psi_0, M U(\theta)\psi_0 \rangle,$$

where $\exp_p$ is the exponential series, and let $\partial_\theta \mathcal{L}$ be the coefficient of θ in the expansion of $\mathcal{L}$ at $\theta = 0$. Then $|\partial_\theta \mathcal{L}|_p \leq p^{-1}$ always. If U_A is drawn from ν , does $\Pr[|\partial_\theta \mathcal{L}|_p = p^{-1}]$ remain bounded away from zero as $n \rightarrow \infty$, and for which pairs (M, A) ?

Three features of the statement are forced rather than chosen. The ensemble is G_N and not the full unitary group of K^N , because the full group is non-compact and carries no Haar probability measure, so posed over it the question has no meaning; this is the first point of the list below. The generator enters as $p\theta A$ because that is exactly what convergence of $\exp_p$ requires. And the benchmark is p^{-1} rather than 1, because every convergent p -adic ansatz confines its one-parameter subgroups to the principal congruence subgroup, so p^{-1} is the largest gradient any such ansatz permits (Remark 2.3).

A heuristic suggests the answer is yes. The normalised gradient $p^{-1}\partial_\theta \mathcal{L}$ lies in $\mathbb{Z}_p$, and a Haar-random element of $\mathbb{Z}_p$ is a unit with probability $1 - p^{-1}$, so one expects $|\partial_\theta \mathcal{L}|_p = p^{-1}$ with probability close to $1 - p^{-1}$, whatever the qubit count. Theorem 5.6 confirms this, uniformly in n , under a splitting hypothesis on the commutator $[U_B^* M U_B, A]$ reduced modulo p (Definition 5.2). The configurations that fail are those in which $U_B^* M U_B$ and A commute modulo p (Corollary 5.9); as U_B varies they form a clopen union of congruence classes of measure $O(p^{-1})$ (Section 6), small but strictly positive, a distinction the complex theory never has to make. Outside the splitting class the question remains open (Section 6.4).

We take the three ingredients in turn.

- **The ensemble is wrong** (Section 3). $U(K^N)$ is non-compact for every $N \geq 2$; we give an explicit unbounded family at $p = 3$ (Proposition 3.1 and Example 3.2). There is therefore no Haar probability measure on the full unitary group, and the plateau question, phrased over it, is ill posed. The compact replacement is the integral unitary group $G_N = U_N(\mathcal{O})$, which contains every circuit built from the currently known realisable p -adic gate sets (Proposition 3.5).

Table 1

Barren plateaus over $\mathbb{C}$ and over $K = \mathbb{Q}_{p,\mu}$. Each row is one ingredient of the argument of [1] and what becomes of it.

	Complex	p -adic
Ensemble	$U(2^n)$, compact	$U(\mathcal{H}_N)$ non-compact; use $G_N = U_N(\mathcal{O})$
Source of moments	unitary t -designs	character sums over $U_N(\mathbb{F}_{p^2})$
Gradient statistic	variance $O(2^{-n})$	$\Pr[\text{maximal}] \geq 1 - p^{-1} - 10p^{1-r}$
Scaling in n	exponential decay	none
Bad set	full measure as $n \rightarrow \infty$	the locus $[\widehat{M}, A] \equiv 0 \pmod{p}$
Role of expressibility	more expressive $\Rightarrow$ flatter [3]	more random $\Rightarrow$ larger gradients

- **The missing combinatorics are not needed** (Section 4). Haar averages over G_N push forward to uniform counting measures on the finite unitary groups $U_N(\mathcal{O}/\mathfrak{m}^k)$ (Lemma 4.1), and Haar-random states push forward to the uniform measure on a finite Hermitian sphere (Lemma 4.2). At level one this turns every question about leading p -adic digits into a character sum over $\mathbb{F}_{p^2}$, which we evaluate in closed form (Lemma 5.3).
- **The conclusion reverses** (Section 5). Theorem 5.6 shows that for an n -qubit circuit whose observable–generator commutator is r -non-degenerate modulo p in the sense of Definition 5.2,

$$\Pr[|\partial_\theta \mathcal{L}|_p = p^{-1}] \geq 1 - \frac{1}{p} - 10p^{1-r},$$

and p^{-1} is the largest absolute value any convergent p -adic ansatz permits (Remark 2.3). The gradient is therefore generically as large as it can be, with a bound that does not depend on n at all.

Table 1 summarises the comparison. Two features of the p -adic answer are worth flagging early because they have no complex analogue. The bound improves as the circuit becomes more expressive, inverting the relationship established in [3]; and the set of bad observable–generator pairs is not a vague “exponentially large region” but the explicit locus $[\widehat{M}, A] \equiv 0 \pmod{p}$, a condition one can check by reducing two integer matrices and multiplying them.

We fix p odd and $K/\mathbb{Q}_p$ unramified throughout, work in finite dimension, and randomise a single parameter in the standard two-block ansatz of [1]. Section 6 states precisely what is and is not established. The load-bearing restriction is the splitting hypothesis of Definition 5.2, which Remark 5.5 shows is not automatic; the example of Example 5.8 shows it is nevertheless satisfied by the most ordinary pair one might write down. The paper is theoretical: no p -adic quantum hardware exists, and, as recalled in Remark 2.2, the p -adic model has no Monte Carlo readout, so the empirical protocol behind barren-plateau plots in the complex setting has no counterpart here. Section 6.3 argues that the result is nonetheless useful, and that its first use is diagnostic and concerns the complex theory: the plateau survives every change we make to the circuit and not the one we make to the field, which locates the hypothesis a mitigation has to attack.

2. Preliminaries

We compress the background to what the proofs use; [15] is the reference for p -adic arithmetic and [4, 5] for the quantum formalism. A reader who works on variational algorithms but not on non-Archimedean analysis needs three facts, and we state them informally before the definitions.

Small means divisible. The p -adic absolute value measures divisibility, not magnitude: $|x|_p = p^{-v_p(x)}$, where $v_p(x)$ counts the factors of p in x . Taking $p = 3$, we have $|18|_3 = 1/9$ because $18 = 2 \cdot 3^2$, while $|1/9|_3 = 9$ and $|2|_3 = 1$. Integers divisible by a high power of p are the small ones.

The scale is discrete. The values taken by $|\cdot|_p$ are the powers $p^{\mathbb{Z}}$, so there is no continuum of small quantities and no notion of a quantity that is merely somewhat small. A gradient is either a unit, or divisible by p , or by p^2 , with nothing in between. This is why “exponential concentration” has to be re-read before it can be asked about: $\text{Var} = O(2^{-n})$ has no p -adic meaning, and the faithful translation is whether the valuation v_p of the gradient grows with n . Question 1.1 asks the sharpest form of this, which is why the answer here is a probability of attaining one specific value rather than a decay rate.

Sums do not grow, and cancellation is a congruence. The ultrametric inequality $|x + y|_p \leq \max\{|x|_p, |y|_p\}$ holds with equality whenever the two terms have different sizes. A sum is therefore as large as its largest term unless the leading digits cancel exactly, and that cancellation is a condition modulo p – a finite, checkable condition rather than a limiting one. This is the structural reason the whole second-moment analysis of Section 4 collapses onto counting in finite groups, where the complex theory needs t -design combinatorics: reduction modulo p carries Haar measure on the integral unitary group to the uniform measure on a finite unitary group, and questions about leading digits become character sums over $\mathbb{F}_{p^2}$.

One expectation does not carry over, and it is a basic one. Generalised probabilities in this formalism lie in $\mathbb{Q}_p$ rather than in $[0, 1]$, so amplitudes have no frequentist reading and cannot be estimated by repeated measurement; readout is algebraic, and Remark 2.2 states what a plateau means once sampling is unavailable.

2.1. Arithmetic

Fix an odd prime p . For $x \in \mathbb{Q}^*$ let $v_p(x)$ be the exponent of p in x and $|x|_p = p^{-v_p(x)}$, with $|0|_p = 0$ and $v_p(0) = +\infty$; $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for this absolute value and $\mathbb{Z}_p = \{|x|_p \leq 1\}$ its ring of integers. The absolute value is *ultrametric*: $|x + y|_p \leq \max\{|x|_p, |y|_p\}$, with equality when $|x|_p \neq |y|_p$. Its value group is $p^{\mathbb{Z}}$, a discrete subgroup of $\mathbb{R}_{>0}$; this discreteness is used repeatedly below and is the single most important difference from the real case.

Let $\mu \in \mathbb{Z}_p^\times$ be a non-square unit and $K = \mathbb{Q}_{p,\mu} = \mathbb{Q}_p(\sqrt{\mu})$, the unramified quadratic extension, with involution $\sigma(x + \sqrt{\mu}y) = x - \sqrt{\mu}y$, written $\bar{z}$. The absolute value extends by $|z|_p = |\text{Nm}(z)|_p^{1/2}$, where

$$\text{Nm} : K \longrightarrow \mathbb{Q}_p, \quad \text{Nm}(z) = z\bar{z}$$

is the field norm. Write $\mathcal{O}$ for the ring of integers of K and $\mathfrak{m} = p\mathcal{O}$ for its maximal ideal, so that

$$\mathcal{O}/\mathfrak{m} = \mathbb{F}_{p^2}, \quad \mathbb{Z}_p/p\mathbb{Z}_p = \mathbb{F}_p,$$

σ reduces to the Frobenius $x \mapsto x^p$ of $\mathbb{F}_{p^2}/\mathbb{F}_p$, and Nm reduces to the norm $z \mapsto z^{1+p}$ of $\mathbb{F}_{p^2}/\mathbb{F}_p$. For an integral z we write $\tilde{z} \in \mathbb{F}_{p^2}$ for its residue, and we apply the same notation entrywise to vectors and matrices. Because $K/\mathbb{Q}_p$ is unramified, $\text{Nm}(K^*)$ is exactly the set of elements of even valuation [16].

We isolate one arithmetic fact, used three times.

Lemma 2.1 (Unit norms). $\text{Nm}(\mathcal{O}^\times) = \mathbb{Z}_p^\times$.

Proof. The inclusion $\subseteq$ is clear. Conversely let $c \in \mathbb{Z}_p^\times$. The norm of $\mathbb{F}_{p^2}/\mathbb{F}_p$ is the map $z \mapsto z^{1+p}$ on $\mathbb{F}_{p^2}^\times$; its kernel has order $p + 1$, so its image has order $(p^2 - 1)/(p + 1) = p - 1$ and is all of $\mathbb{F}_p^\times$. Pick $s_1 \in \mathcal{O}^\times$ with $\text{Nm}(s_1) \equiv c \pmod{p}$. Then $c/\text{Nm}(s_1) \in 1 + p\mathbb{Z}_p$, and every element of $1 + p\mathbb{Z}_p$ is a square in $\mathbb{Z}_p$ for p odd, by Hensel’s lemma applied to $X^2 - c/\text{Nm}(s_1)$ at $X = 1$. Writing $c/\text{Nm}(s_1) = X^2$ with $X \in 1 + p\mathbb{Z}_p$ and setting $s = Xs_1$ gives $\text{Nm}(s) = X^2\text{Nm}(s_1) = c$. $\square$

2.2. States, gates, gradients

Let $\mathcal{H}_N = K^N$ with $\|v\| = \max_i |v_i|_p$ and $\langle v, w \rangle = \sum_i \bar{v}_i w_i$, linear in the second argument. This is the model finite-dimensional p -adic Hilbert space [4]; the n -qubit space is $\mathcal{H}_N$ with

$N = 2^n$, built from $\mathcal{H}_2^{\otimes n}$ via the p -adic tensor product [12]. An operator $A \in \mathcal{B}(\mathcal{H}_N)$ has $\|A\| = \max_{ij} |A_{ij}|_p$ and adjoint $(A^*)_{ij} = \overline{A_{ji}}$; ultrametricity makes $\|\cdot\|$ submultiplicative, since $|(AB)_{ij}|_p = |\sum_k A_{ik} B_{kj}|_p \leq \max_k |A_{ik}|_p |B_{kj}|_p$. A *gate* is a U with $U^* U = \mathbf{1}$; these form the group $U(\mathcal{H}_N)$. A vector ψ is *admissible* if $\langle \psi, \psi \rangle = 1$, and observables are selfadjoint operators.

Two features of the model matter below. First, positivity is unavailable: $\langle \psi, \psi \rangle \geq 0$ is meaningless because K is unordered, so states form an affine rather than a convex set and measurement is described by selfadjoint operator-valued measures with $\mathbb{Q}_p$ -valued generalised probabilities [4, 17]. Second:

Remark 2.2 (No Monte Carlo readout [4]). Generalised probabilities lie in $\mathbb{Q}_p$, not $[0, 1]$, and cannot be estimated by repeated sampling. Readout is algebraic: the algorithm returns p -adic numbers which the user inspects digit by digit. The operational meaning of a barren plateau changes accordingly. In the complex setting a small gradient is bad because it takes many shots to resolve; in the p -adic setting there are no shots, and a gradient of high valuation is bad because the leading digits of the parameter update vanish.

For the one-parameter dependence we use the exponential of the p -adic Lie algebra. The series $\exp_p(X) = \sum_{m \geq 0} X^m / m!$ converges whenever $\|X\| < p^{-1/(p-1)}$ [18], this being the standard bound we adopt, and for $A^* = -A$ it defines a unitary. Following [1] we take the two-block ansatz, in the normalisation forced by Remark 2.3 below:

$$U(\theta) = U_B \exp_p(p\theta A) U_A, \quad \theta \in \mathbb{Z}_p, \quad A^* = -A, \quad A \in \mathcal{B}(\mathcal{H}_N) \text{ integral.} \quad (1)$$

Since $\mathcal{L}$ is p -adic analytic in θ , we define $\partial_\theta \mathcal{L}$ as the coefficient of θ in its Taylor expansion at $\theta = 0$.

Remark 2.3 (The scale p is forced, not chosen). For p odd we have $0 < \frac{1}{p-1} < 1$, and the value group of $\|\cdot\|$ is the *discrete* set $p^{\mathbb{Z}}$. No operator norm lies strictly between p^{-1} and 1, so the convergence bound $\|X\| < p^{-1/(p-1)}$ is not a mild constraint that could be relaxed by a better estimate: it is *equivalent* to $\|X\| \leq p^{-1}$, that is, to $X \equiv 0 \pmod{\mathfrak{m}}$. Under it, every term $X^m / m!$ with $m \geq 1$ has norm at most $p^{-m} p^{(m-1)/(p-1)} \leq p^{-1}$, since $v_p(m!) \leq (m-1)/(p-1)$, so

$$\exp_p(\theta X) \equiv \mathbf{1} \pmod{\mathfrak{m}} \quad \text{for all } \theta \in \mathbb{Z}_p.$$

The one-parameter subgroup thus lies inside the principal congruence subgroup $\ker(\pi_1)$ of Section 4, and every Taylor coefficient of $\mathcal{L}$ of positive order has valuation at least 1. In particular

$$|\partial_\theta \mathcal{L}|_p \leq p^{-1}$$

for any integral observable and any ansatz meeting the convergence bound. The generator in (1) is written pA with A integral for exactly this reason: the factor p is not a normalisation we are free to choose, it is the entire content of the convergence condition, and p^{-1} rather than 1 is the correct benchmark for “as large as possible”. We write

$$\nabla \mathcal{L} := p^{-1} \partial_\theta \mathcal{L} \quad (2)$$

for the *normalised gradient*, which does satisfy $|\nabla \mathcal{L}|_p \leq 1$, and state our results in terms of it.

3. The ensemble: the unitary group is not compact

The barren-plateau question presupposes a Haar *probability* measure on the group of n -qubit unitaries. Over $\mathbb{C}$ this is automatic because $U(2^n)$ is compact. Over K it fails, and it fails badly.

Proposition 3.1. *For every $N \geq 2$ the group $U(\mathcal{H}_N)$ is unbounded in the operator norm, hence non-compact, and carries no Haar probability measure.*

Proof. It suffices to treat $N = 2$ and pad with the identity, since $U \mapsto U \oplus \mathbf{1}_{N-2}$ embeds $U(\mathcal{H}_2)$ isometrically in $U(\mathcal{H}_N)$. We construct, for each $m \geq 1$, a unitary of operator norm p^m .

Put $a = p^{-m}$ and $t = \text{Nm}(a) = p^{-2m}$. Since $|t|_p = p^{2m} > 1 = |1|_p$, ultrametricity gives $|1 - t|_p = |t|_p$, so $v_p(1 - t) = -2m$ is even. Because $K/\mathbb{Q}_p$ is unramified, $\text{Nm}(K^*)$ consists exactly of the elements of even valuation, so there is $c \in K$ with $\text{Nm}(c) = 1 - t$. The vector $u = (a, c)^\top$ then satisfies

$$\langle u, u \rangle = \bar{a}a + \bar{c}c = \text{Nm}(a) + \text{Nm}(c) = t + (1 - t) = 1,$$

so u is admissible, while $|a|_p = p^m$.

Both e_1 and u are admissible, so $e_1 \mapsto u$ is an isometry between one-dimensional subspaces of the non-degenerate Hermitian space $\mathcal{H}_2$. Witt's extension theorem for Hermitian forms over a field with involution, in characteristic $\neq 2$ [19, Ch. 7], extends it to an isometry U of $\mathcal{H}_2$, that is, to an element of $U(\mathcal{H}_2)$. Its first column is u , so $\|U\| \geq |a|_p = p^m$.

Finally, $U(\mathcal{H}_N)$ is a closed subgroup of $\text{GL}_N(K)$, hence locally compact; the compact subsets of $K^{N \times N}$ are the closed bounded ones, so an unbounded closed subgroup is not compact; and a locally compact group admits a finite Haar measure only when it is compact. $\square$

Example 3.2 ($p = 3$). Take $\mu = -1$, a non-square unit modulo 3, so that $K = \mathbb{Q}_3(i)$ is unramified and $\text{Nm}(x + iy) = x^2 + y^2$. With $m = 1$ we need $\text{Nm}(a) = 1/9$ and $\text{Nm}(c) = 8/9$. Take $a = 1/3$ and $c = \frac{1}{3}(x_1 + iy_1)$ with $x_1^2 + y_1^2 = 8$: the congruence $x_1^2 + y_1^2 \equiv 2 \pmod{3}$ has the solution $x_1 \equiv y_1 \equiv 1$, which lifts to $\mathbb{Z}_p$ by Hensel's lemma applied to $X^2 - 7$ at $X = 1$, whose derivative $2X$ is a unit there. Then $|a|_p = |c|_p = 3$, and completing $(a, c)^\top$ to a unitary gives an element of $U(\mathcal{H}_2)$ of operator norm 3. Replacing $1/3$ by 3^{-m} gives operator norm 3^m .

Proposition 3.1 is not an artefact of a poor choice of Hermitian form. Over a local field the standard form $\sum \bar{z}_i z_i$ is isotropic in every dimension $N \geq 2$ – for $N = 2$ isotropy is exactly the statement $-1 \in \text{Nm}(K^*)$, which holds in the unramified case – and unitary groups of isotropic Hermitian forms are non-compact [20, Ch. 3]. The compactness that the complex theory takes for granted comes from the positive-definiteness of $\sum \bar{z}_i z_i$ over $\mathbb{C}$, and positivity is precisely what the p -adic model gives up.

Definition 3.3 (Integral unitary group). $G_N = \text{U}_N(\mathcal{O}) = \{U \in U(\mathcal{H}_N) : U_{ij} \in \mathcal{O} \text{ for all } i, j\}$.

Proposition 3.4. G_N is a compact group and carries a unique Haar probability measure ν .

Proof. G_N is closed under products, and under inverses because $U^{-1} = U^*$ has entries $\overline{U_{ji}} \in \mathcal{O}$. As a subset of $\mathcal{O}^{N \times N}$ it is bounded, and it is closed, being the solution set of the polynomial system $U^*U = \mathbf{1}$ over the compact ring $\mathcal{O}$. A closed bounded subset of $\mathcal{O}^{N^2}$ is compact, and a compact group has a unique normalised Haar measure. $\square$

Restricting to G_N costs nothing for quantum computing, because everything one can currently build already lies there.

Proposition 3.5. Every gate whose matrix entries lie in $\mathcal{O}$ generates, under composition and tensor product, a subgroup of G_N . This applies to the p -adic Pauli gates and CNOT of [4], to the gate set drawn from irreducible representations of $\text{SO}(3)_p$ of [13], and when $\sqrt{2} \in K$ to the p -adic Hadamard gate.

Proof. Integrality is preserved by products and tensor products, and G_N is a group by Proposition 3.4. For the three gate sets it remains only to observe that their entries are integral. The Pauli and CNOT entries lie in $\{0, \pm 1\}$. The entries of the $\text{SO}(3)_p$ gates are roots of unity in K and zeros; roots of unity are units, hence integral. For the Hadamard gate, if $\sqrt{2} \in K$ then $|\sqrt{2}|_p = |2|_p^{1/2} = 1$ for odd p , so $1/\sqrt{2} \in \mathcal{O}^\times$ and all four entries are units. $\square$

Remark 3.6. The contrast with $\mathbb{C}$ runs deeper than a change of group. There, “Haar-random unitary” and “deep random circuit” converge to each other, and that convergence is what makes the t -design argument of [1] bite. Over K the two notions come apart at the outset: the group of all unitaries is too large to randomise over, and the group generated by realisable gates is a compact subgroup with a different measure-theoretic structure altogether. We take G_N as the ensemble because it is the smallest natural compact group containing all realisable circuits. Sharper results should be available for the compact group actually generated by a fixed gate set, which is where a hardware-aware analysis would have to go.

4. Haar averages are finite counting problems

The argument of [1] rests on unitary t -design combinatorics, for which we know of no p -adic counterpart. They are not needed. Compactness of G_N arrives together with a filtration that turns every question about the leading digits of a Haar average into a finite count.

For $k \geq 1$ write $\mathcal{O}_k = \mathcal{O}/\mathfrak{m}^k$ and let $\pi_k : G_N \rightarrow \mathrm{U}_N(\mathcal{O}_k)$ be reduction of matrix entries modulo $\mathfrak{m}^k$, where $\mathrm{U}_N(\mathcal{O}_k)$ is the unitary group of the reduced Hermitian form over the finite ring $\mathcal{O}_k$. For $k = 1$ this is the finite unitary group $\mathrm{U}_N(\mathbb{F}_{p^2})$.

Lemma 4.1 (Pushforward). *For every $k \geq 1$ the map π_k is a surjective group homomorphism and $(\pi_k)_*\nu$ is the uniform probability measure on $\mathrm{U}_N(\mathcal{O}_k)$. Consequently, if $F : G_N \rightarrow \mathbb{Z}_p$ is a function whose residue $F \bmod p^k$ depends on U only through $\pi_k(U)$, then*

$$\nu(\{U : v_p(F(U)) \geq k\}) = \frac{\#\{g \in \mathrm{U}_N(\mathcal{O}_k) : F(g) \equiv 0\}}{|\mathrm{U}_N(\mathcal{O}_k)|}.$$

Proof. Reduction modulo $\mathfrak{m}^k$ is a ring homomorphism $\mathcal{O} \rightarrow \mathcal{O}_k$ commuting with σ , because $\sigma(\mathfrak{m}^k) = \mathfrak{m}^k$; it therefore carries the relation $U^*U = \mathbf{1}$ to its reduction, and π_k is a homomorphism.

For surjectivity, take $g \in \mathrm{U}_N(\mathcal{O}_k)$ and lift its entries arbitrarily to a matrix $g_0 \in \mathcal{B}(\mathcal{H}_N)$ with integral entries. Then $g_0^*g_0$ is selfadjoint and congruent to $\mathbf{1}$ modulo $\mathfrak{m}^k$, say

$$g_0^*g_0 = \mathbf{1} + p^l S, \quad l = k \geq 1, \quad S^* = S \text{ integral}.$$

We improve l by Newton iteration. Set $T = -S/2$, which is integral because p is odd, and selfadjoint because S is; put $g_1 = g_0(\mathbf{1} + p^l T)$. Expanding, and using $T^* = T$ in the leftmost factor,

$$g_1^*g_1 = (\mathbf{1} + p^l T)(\mathbf{1} + p^l S)(\mathbf{1} + p^l T) = \mathbf{1} + p^l(T + S + T) + p^{2l}(TS + ST + T^2) + p^{3l}TST,$$

and $T + T = -S$ kills the order- p^l term, so $g_1^*g_1 = \mathbf{1} + p^{2l}S'$ with S' selfadjoint and integral. Moreover $g_1 \equiv g_0 \bmod \mathfrak{m}^l$. Iterating produces a sequence $g_0, g_1, g_2, \dots$ with $g_{j+1} \equiv g_j \bmod \mathfrak{m}^{2^j k}$, which is Cauchy in the complete space $\mathcal{B}(\mathcal{H}_N)$ of integral matrices; its limit h is integral, satisfies $h^*h = \mathbf{1}$ hence is invertible with $h^{-1} = h^*$ integral, so $h \in G_N$ and satisfies $h \equiv g_0 \bmod \mathfrak{m}^k$, that is $\pi_k(h) = g$.

Being a continuous surjective homomorphism from the compact group G_N onto the finite group $\mathrm{U}_N(\mathcal{O}_k)$, π_k carries ν to a translation-invariant probability measure on the target, which by uniqueness of Haar measure is the uniform one. The displayed identity is the definition of the pushforward applied to the indicator of $\{F \equiv 0 \bmod p^k\}$, which by hypothesis is a π_k -pullback. $\square$

We shall also need that Haar measure on G_N moves a fixed admissible vector to the uniform measure on the integral sphere

$$\mathbb{S}_N = \{u \in \mathcal{O}^N : \langle u, u \rangle = 1\}, \quad \tilde{\mathbb{S}}_N = \{\tilde{u} \in \mathbb{F}_{p^2}^N : \langle \tilde{u}, \tilde{u} \rangle = 1\}.$$

The proof below is elementary and self-contained; it replaces an appeal to smoothness of the unitary group scheme, and it makes visible exactly where p odd is used.

Lemma 4.2 (Transitivity). *Let $N \geq 1$. Then G_N acts transitively on $\mathbb{S}_N$, and for $\psi_0 \in \mathbb{S}_N$ the pushforward of ν under $U \mapsto U\psi_0$ is the unique G_N -invariant probability measure on $\mathbb{S}_N$. Reduction modulo $\mathfrak{m}$ maps $\mathbb{S}_N$ onto $\widetilde{\mathbb{S}}_N$ and carries that measure to the uniform measure on $\widetilde{\mathbb{S}}_N$.*

Proof. We argue in four steps.

(i) *Every $u \in \mathbb{S}_N$ is unimodular.* From $1 = |\langle u, u \rangle|_p \leq \max_i |u_i|_p^2 \leq 1$ we get $\max_i |u_i|_p = 1$.

(ii) *A norm-one vector splits off an orthogonal complement.* Let L be any free Hermitian $\mathcal{O}$ -lattice whose Gram determinant is a unit call such a lattice *unimodular* and let $u \in L$ with $\langle u, u \rangle = 1$. The map $P(x) = x - \langle u, x \rangle u$ is $\mathcal{O}$ -linear, because $\langle u, \cdot \rangle$ is linear in its second argument, and $\langle u, P(x) \rangle = \langle u, x \rangle (1 - \langle u, u \rangle) = 0$. Hence $L = \mathcal{O}u \oplus u^\perp$ orthogonally, where $u^\perp = \ker \langle u, \cdot \rangle$. As a direct summand of a free module over the local ring $\mathcal{O}$, the lattice $u^\perp$ is free, of rank one less. It is again unimodular: the change of basis from a basis of L to one adapted to the splitting multiplies the Gram determinant by $\text{Nm}(\det Q)$ for some $Q \in \text{GL}(\mathcal{O})$, hence by a unit, while the adapted Gram matrix is $\text{diag}(1, G)$, so $\det G \in \mathcal{O}^\times$. Applied to $L = \mathcal{O}^N$ with the standard form, whose Gram determinant is 1, this splits $\mathcal{O}^N = \mathcal{O}u \perp u^\perp$ with $u^\perp$ unimodular of rank $N - 1$.

(iii) *Every unimodular Hermitian $\mathcal{O}$ -lattice L of rank ≥ 1 contains a vector of norm 1.* The reduction $L/\mathfrak{m}L$ carries a non-degenerate Hermitian form over $\mathbb{F}_{p^2}$. That form cannot vanish on the diagonal: if $\langle x, x \rangle = 0$ for all x , then polarising gives $\text{Tr}_{\mathbb{F}_{p^2}/\mathbb{F}_p} \langle x, y \rangle = 0$ for all x, y , and replacing y by $\sqrt{\mu}y$ gives $\text{Tr}_{\mathbb{F}_{p^2}/\mathbb{F}_p} (\sqrt{\mu} \langle x, y \rangle) = 0$; writing $\langle x, y \rangle = a + b\sqrt{\mu}$ these two conditions read $2a = 0$ and $2b\mu = 0$, so $\langle x, y \rangle = 0$ identically, contradicting non-degeneracy. Choose therefore $x_0 \in L$ whose residue has non-zero norm, so that $c = \langle x_0, x_0 \rangle \in \mathbb{Z}_p^\times$. By Lemma 2.1 there is $s \in \mathcal{O}^\times$ with $\text{Nm}(s) = c^{-1}$, and then $\langle sx_0, sx_0 \rangle = \text{Nm}(s)c = 1$.

(iv) *Transitivity.* Any two unimodular Hermitian $\mathcal{O}$ -lattices of equal rank are isometric, by induction on the rank: each contains a vector of norm 1 by (iii), and (ii) splits that vector off leaving unimodular lattices of rank one less, the rank-zero case being trivial. Given $u, u' \in \mathbb{S}_N$, apply (ii) to both and choose an isometry $\phi : u^\perp \rightarrow u'^\perp$; the map sending $u \mapsto u'$ and acting as ϕ on $u^\perp$ is an isometry of $\mathcal{O}^N$, that is an element $U \in G_N$ with $Uu = u'$.

Uniqueness of the invariant probability measure on a transitive compact group action is standard. For the reduction, note first that every $\tilde{u} \in \widetilde{\mathbb{S}}_N$ lifts: choose any integral u_0 with residue $\tilde{u}$, so $c = \langle u_0, u_0 \rangle \in 1 + p\mathbb{Z}_p$; by Hensel $c^{-1} = X^2$ with $X \in 1 + p\mathbb{Z}_p$, and $u = Xu_0$ lies in $\mathbb{S}_N$ with the same residue. Reduction therefore maps $\mathbb{S}_N$ onto $\widetilde{\mathbb{S}}_N$, and it intertwines the action of G_N with the action of $U_N(\mathbb{F}_{p^2})$, which is surjective by Lemma 4.1. The image of an invariant measure under an equivariant surjection onto a transitive finite action is the uniform measure. $\square$

5. Gradient valuations

5.1. The gradient is a Hermitian form

Lemma 5.1. *For the ansatz (1) with $\psi = U_A\psi_0$ and $\widehat{M} = U_B^*MU_B$, the normalised gradient (2) is*

$$\nabla \mathcal{L} = \langle \psi, C\psi \rangle, \quad C = [\widehat{M}, A] = \widehat{M}A - A\widehat{M}.$$

The operator C is selfadjoint, so $\nabla \mathcal{L} \in \mathbb{Q}_p$; and if M, A, U_A, U_B, ψ_0 are integral then $\nabla \mathcal{L} \in \mathbb{Z}_p$, so $|\nabla \mathcal{L}|_p \leq 1$, equivalently $|\partial_\theta \mathcal{L}|_p \leq p^{-1}$.

Proof. Since $U(\theta)\psi_0 = U_B \exp_p(p\theta A)\psi$ and $\langle U_B x, MU_B x \rangle = \langle x, \widehat{M}x \rangle$, we have $\mathcal{L}(\theta) = \langle \exp_p(p\theta A)\psi, \widehat{M} \exp_p(p\theta A)\psi \rangle$. The coefficient of θ in the expansion is

$$\langle pA\psi, \widehat{M}\psi \rangle + \langle \psi, \widehat{M}pA\psi \rangle = p \langle \psi, (A^*\widehat{M} + \widehat{M}A)\psi \rangle,$$

and $A^* = -A$ turns the bracket into $C = [\widehat{M}, A]$; dividing by p gives the stated formula. For selfadjointness, $C^* = (\widehat{MA})^* - (A\widehat{M})^* = A^*\widehat{M} - \widehat{MA}^* = \widehat{MA} - A\widehat{M} = C$. For a selfadjoint C one has $\langle \psi, C\psi \rangle = \langle C\psi, \psi \rangle = \langle \psi, C^*\psi \rangle = \langle \psi, C\psi \rangle$, so the scalar is fixed by the involution and lies in the fixed field $\mathbb{Q}_p$. Integrality is closure of $\mathcal{O}$ under sums and products, together with $\mathcal{O} \cap \mathbb{Q}_p = \mathbb{Z}_p$. $\square$

So the normalised gradient is a Hermitian form evaluated at a point of the sphere. Both of its arguments are Haar-random: ψ is uniform on $\mathbb{S}_N$ by Lemma 4.2, and C is determined by the block U_B . What remains is a finite count, which we now do by hand.

5.2. The counting lemma

Fix $N \geq 2$ and a selfadjoint Γ over $\mathbb{F}_{p^2}$. For $c \in \mathbb{F}_p$ put

$$\mathcal{N}_c = \#\{u \in \mathbb{F}_{p^2}^N : \langle u, u \rangle = 1, \langle u, \Gamma u \rangle = c\}.$$

Definition 5.2 (Non-degeneracy). Say Γ is *r-non-degenerate* if it admits an orthonormal eigenbasis over $\mathbb{F}_{p^2}$ with all eigenvalues in $\mathbb{F}_p$, and no eigenvalue has multiplicity larger than $N - r$. Writing $m_{\max}$ for the largest eigenvalue multiplicity, the second condition reads $r \leq N - m_{\max}$.

Lemma 5.3. *If Γ is r-non-degenerate then for every $c \in \mathbb{F}_p$*

$$\left| \frac{\mathcal{N}_c}{|\widetilde{\mathbb{S}}_N|} - \frac{1}{p} \right| \leq 10p^{1-r}, \quad \text{and} \quad |\widetilde{\mathbb{S}}_N| = p^{2N-1} - (-1)^N p^{N-1}.$$

Proof. Work in an orthonormal eigenbasis of Γ , with eigenvalues $\lambda_1, \dots, \lambda_N \in \mathbb{F}_p$. The two constraints read $\sum_i \text{Nm}(u_i) = 1$ and $\sum_i \lambda_i \text{Nm}(u_i) = c$, where now $\text{Nm} : \mathbb{F}_{p^2} \rightarrow \mathbb{F}_p$, $\text{Nm}(z) = z^{1+p}$. The fibres of Nm have sizes $w(0) = 1$ and $w(s) = p + 1$ for $s \neq 0$, since Nm restricts to a surjection $\mathbb{F}_{p^2}^\times \rightarrow \mathbb{F}_p^\times$ with kernel of order $p + 1$. Substituting $t_i = \text{Nm}(u_i)$,

$$\mathcal{N}_c = \sum_{\substack{t \in \mathbb{F}_p^N \\ \sum t_i = 1, \sum \lambda_i t_i = c}} \prod_{i=1}^N w(t_i).$$

Let χ be a non-trivial additive character of $\mathbb{F}_p$ and detect the two constraints by $\frac{1}{p} \sum_\alpha \chi(\alpha(\cdot))$. Then

$$\mathcal{N}_c = \frac{1}{p^2} \sum_{\alpha, \beta \in \mathbb{F}_p} \chi(-\alpha - \beta c) \prod_{i=1}^N T(\alpha + \beta \lambda_i), \quad T(s) = \sum_{t \in \mathbb{F}_p} w(t) \chi(st).$$

Direct evaluation gives $T(0) = 1 + (p + 1)(p - 1) = p^2$ and, for $s \neq 0$, $T(s) = 1 + (p + 1) \sum_{t \neq 0} \chi(st) = 1 - (p + 1) = -p$. Hence, writing $z(\alpha, \beta) = \#\{i : \alpha + \beta \lambda_i = 0\}$,

$$\prod_i T(\alpha + \beta \lambda_i) = (p^2)^z (-p)^{N-z} = (-1)^{N-z} p^{N+z}.$$

Split the double sum into three ranges.

Case $\alpha = \beta = 0$. Here $z = N$ and the term is p^{2N} .

Case $\beta = 0, \alpha \neq 0$. Here $z = 0$, and $\sum_{\alpha \neq 0} \chi(-\alpha) = -1$; the total contribution is $-(-1)^N p^N$.

Case $\beta \neq 0$. For fixed $\beta \neq 0$ the substitution $\lambda = -\alpha/\beta$ is a bijection on α . It gives $\alpha + \beta \lambda_i = \beta(\lambda_i - \lambda)$, so $z = m(\lambda)$, the multiplicity of λ in the spectrum, zero if λ is not an eigenvalue; and $-\alpha - \beta c = \beta(\lambda - c)$. Summing over $\beta \neq 0$ gives $\sum_{\beta \neq 0} \chi(\beta(\lambda - c)) = p\delta_{\lambda, c} - 1$, so this case contributes $\Sigma(c) = \sum_{\lambda \in \mathbb{F}_p} (-1)^{N-m(\lambda)} p^{N+m(\lambda)} (p\delta_{\lambda, c} - 1)$. Collecting,

$$p^2 \mathcal{N}_c = p^{2N} - (-1)^N p^N + \Sigma(c). \quad (3)$$

We first extract the sphere count. Summing (3) over $c \in \mathbb{F}_p$ gives $\sum_c \mathcal{N}_c = |\tilde{\mathbb{S}}_N|$ on the left, while on the right $\sum_c \Sigma(c) = \sum_\lambda (-1)^{N-m(\lambda)} p^{N+m(\lambda)} (p-p) = 0$; therefore $p^2 |\tilde{\mathbb{S}}_N| = p^{2N+1} - (-1)^N p^{N+1}$, which is the stated formula. In particular $|\tilde{\mathbb{S}}_N| \geq p^{2N-1} - p^{N-1} \geq \frac{1}{2} p^{2N-1}$ for $N \geq 2$, and $|\tilde{\mathbb{S}}_N|/p = p^{2N-2} - (-1)^N p^{N-2}$.

Now bound $\Sigma(c)$. At most p values of λ have $m(\lambda) = 0$; for those the summand has absolute value at most $p^N \cdot p = p^{N+1}$ when $\lambda = c$ and at most p^N otherwise, contributing at most $p^{N+1} + (p-1)p^N \leq 2p^{N+1}$ in total. At most p values of λ have $m(\lambda) \geq 1$, and for those the summand is at most $p^{N+m_{\max}} \cdot p$ in absolute value, contributing at most $p^{N+m_{\max}+2}$. Hence, using $m_{\max} \geq 1$,

$$|p^2 \mathcal{N}_c - p^{2N}| \leq p^N + 2p^{N+1} + p^{N+m_{\max}+2} \leq 4p^{N+m_{\max}+2},$$

that is $\mathcal{N}_c = p^{2N-2} + E$ with $|E| \leq 4p^{N+m_{\max}}$. Combining with $|\tilde{\mathbb{S}}_N|/p = p^{2N-2} - (-1)^N p^{N-2}$,

$$\left| \mathcal{N}_c - \frac{|\tilde{\mathbb{S}}_N|}{p} \right| \leq 4p^{N+m_{\max}} + p^{N-2} \leq 5p^{N+m_{\max}},$$

and dividing by $|\tilde{\mathbb{S}}_N| \geq \frac{1}{2} p^{2N-1}$ gives $10p^{m_{\max}-N+1} \leq 10p^{1-r}$. $\square$

Remark 5.4 (When the bound bites). Lemma 5.3 is vacuous for small r : the deviation bound $10p^{1-r}$ exceeds 1 unless $p^{r-1} > 10$. Combined with the leading term $1/p$ in Theorem 5.6, the conclusion is non-trivial as soon as $p^{r-1} > 10p/(p-1)$, which holds for $r \geq 4$ when $p = 3$, for $r \geq 3$ when $p \geq 11$, and for $r \geq 2$ when $p \geq 13$. In the regime of interest r grows like 2^n (Example 5.8 and Remark 5.11), so the condition is met from three qubits onward for every odd p .

Remark 5.5 (The splitting clause is a real restriction). It is tempting to read the first clause of Definition 5.2 as an analogue of the complex spectral theorem, hence automatic. It is not. Over $\mathbb{F}_{q^2}$ a Hermitian matrix can have eigenvalues outside the fixed field $\mathbb{F}_q$. Indeed if $\Gamma v = \lambda v$ then

$$\lambda \langle v, v \rangle = \langle v, \Gamma v \rangle = \langle \Gamma v, v \rangle = \bar{\lambda} \langle v, v \rangle,$$

so $\lambda \neq \bar{\lambda}$ forces $\langle v, v \rangle = 0$; the same computation applied to two vectors of the same eigenspace shows that eigenspace is totally isotropic, and no orthonormal eigenbasis can exist. Concretely, take $q = 3$ and $\mathbb{F}_9 = \mathbb{F}_3(\iota)$ with $\iota^2 = -1$, so that the involution is $x + \iota y \mapsto x - \iota y$ and $\text{Nm}(x + \iota y) = x^2 + y^2$. With $b = 1 + \iota$, which has $\text{Nm}(b) = 2$, the matrix

$$\Gamma = \begin{pmatrix} 0 & b \\ \bar{b} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 + \iota \\ 1 - \iota & 0 \end{pmatrix}$$

is Hermitian with characteristic polynomial $x^2 - \text{Nm}(b) = x^2 + 1$, whose roots $\pm \iota$ lie in $\mathbb{F}_9 \setminus \mathbb{F}_3$. Its ι -eigenvector is $(1 + \iota, \iota)^\top$, of norm $2 + 1 = 0$.

So Definition 5.2 restricts to those Γ whose characteristic polynomial splits over $\mathbb{F}_p$ with an orthonormal eigenbasis, and Theorem 5.6 is conditional on that. The general case needs the classification of pencils of Hermitian forms over a finite field rather than a single diagonalisation; we have not carried it out, and we do not know whether the conclusion of Theorem 5.6 survives without the hypothesis. This is the principal gap in the paper. What we can say is that the restricted class is not exotic: Example 5.8 exhibits a textbook observable generator pair inside it, with r as large as the pigeonhole bound of Remark 5.11 allows.

5.3. Main result

Theorem 5.6 (No barren plateaus). *Let p be odd, $K/\mathbb{Q}_p$ unramified, $N = 2^n$. Fix*

- an integral observable $M = M^*$ and an integral generator $A = -A^*$, entering the ansatz (1) as $\exp_p(p\theta A)$;
- an admissible integral ψ_0 ;
- an arbitrary $U_B \in G_N$, and write $\widehat{M} = U_B^* M U_B$, $C = [\widehat{M}, A]$ and $\Gamma = \widetilde{C}$;

and suppose Γ is r -non-degenerate. Let U_A be drawn from the Haar measure ν on G_N . Then

$$\Pr\left[|\partial_\theta \mathcal{L}|_p = p^{-1}\right] = \Pr\left[|\nabla \mathcal{L}|_p = 1\right] \geq 1 - \frac{1}{p} - 10p^{1-r}.$$

That is, the gradient attains the largest absolute value any convergent p -adic ansatz permits (Remark 2.3), with a probability bounded below independently of the number of qubits n .

Proof. By Lemma 5.1, $\nabla \mathcal{L} = \langle \psi, C\psi \rangle \in \mathbb{Z}_p$ with $\psi = U_A \psi_0$, so $|\nabla \mathcal{L}|_p \leq 1$ with equality precisely when $v_p(\nabla \mathcal{L}) = 0$, that is when the residue $\langle \tilde{\psi}, \Gamma \tilde{\psi} \rangle \in \mathbb{F}_p$ is non-zero.

The residue $\langle \psi, C\psi \rangle \bmod p$ depends on ψ only through $\tilde{\psi}$, and Γ is fixed by hypothesis. By Lemma 4.2, $\psi = U_A \psi_0$ is distributed according to the invariant measure on $\mathbb{S}_N$ as U_A ranges over G_N , so $\tilde{\psi}$ is uniform on $\widetilde{\mathbb{S}}_N$. Applying Lemma 5.3 with $c = 0$,

$$\Pr[v_p(\nabla \mathcal{L}) \geq 1] = \frac{\mathcal{N}_0}{|\widetilde{\mathbb{S}}_N|} \leq \frac{1}{p} + 10p^{1-r},$$

and taking complements gives the claim. The identification of the two events in the statement is $\nabla \mathcal{L} = p^{-1} \partial_\theta \mathcal{L}$. $\square$

Corollary 5.7 (Random second block). *If instead U_A and U_B are drawn independently from ν and Γ is r -non-degenerate for ν -almost every U_B , the same bound holds.*

Proof. The bound of Theorem 5.6 is uniform over the admissible U_B , so it survives averaging over U_B . $\square$

Because Theorem 5.6 takes a concrete pair (M, A) as input, its hypothesis is something one can check rather than assume. Here is a pair that passes.

Example 5.8 (A witness with $r = 2^{n-1}$). Let $p \equiv 3 \pmod{4}$, so that $\mu = -1$ is a non-square unit and $K = \mathbb{Q}_p(i)$ with $i^2 = -1$, $\bar{i} = -i$. On n qubits take the observable and generator

$$M = X_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad A = -\frac{i}{2}(\mathbf{1} - Z_1) = -i \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

blocks of size 2^{n-1} ; both are integral, since $2 \in \mathbb{Z}_p^\times$ for p odd, and $M^* = M$ while $A^* = -A$ because $\bar{i} = -i$. In words: measure Pauli X on the first qubit and rotate the first qubit about Z . Take $U_B = \mathbf{1}$ and $\psi_0 = e_1$. Then

$$C = [M, A] = \begin{pmatrix} 0 & -i\mathbf{1} \\ i\mathbf{1} & 0 \end{pmatrix} = Y_1,$$

the p -adic Pauli Y on the first qubit, and $\Gamma = \widetilde{C}$ satisfies $\Gamma^2 = \mathbf{1}$. Its eigenvalues are therefore $\pm 1 \in \mathbb{F}_p$, and since Γ has trace 0 each occurs with multiplicity 2^{n-1} . The eigenspaces are $E_\pm = \{(x, \pm ix)^\top\}$ and carry the non-degenerate forms $\langle (x, \pm ix), (z, \pm iz) \rangle = 2\langle x, z \rangle$, so each has an orthonormal basis and Γ is unitarily diagonalisable. Hence $m_{\max} = 2^{n-1}$ and Γ is r -non-degenerate with

$$r = N - m_{\max} = 2^{n-1},$$

so Theorem 5.6 gives $\Pr[|\partial_\theta \mathcal{L}|_p = p^{-1}] \geq 1 - p^{-1} - 10p^{1-2^{n-1}}$. At $n = 10$ and $p = 3$ the correction term is smaller than 3^{-500} .

Corollary 5.9. *Assume Γ admits an orthonormal eigenbasis with eigenvalues in $\mathbb{F}_p$. Then Γ fails to be 1-non-degenerate if and only if $\widehat{M}$ and A commute modulo p .*

Proof. Failure of 1-non-degeneracy means $m_{\max} = N$, that is $\Gamma = \lambda \mathbf{1}$ for some $\lambda \in \mathbb{F}_p$. Being a commutator, Γ has trace zero, so $N\lambda = 0$ in $\mathbb{F}_p$; since $N = 2^n$ and p is odd, $p \nmid N$ and therefore $\lambda = 0$. Thus $\Gamma = 0$, which is exactly $[\widehat{M}, A] \equiv 0 \pmod{\mathfrak{m}}$. The converse is immediate. $\square$

Corollary 5.10 (Contrast with the complex case). *For a complex n -qubit circuit whose blocks form 2-designs, $\Pr[|\partial_\theta \mathcal{L}| \geq \varepsilon] \leq O(2^{-n}\varepsilon^{-2})$ by Chebyshev’s inequality applied to the mean-zero, variance- $O(2^{-n})$ gradient of [1]. Over K , by Theorem 5.6, the gradient attains its maximal attainable absolute value with probability bounded below uniformly in n .*

Remark 5.11 (How large can r be, and how large is it typically). Two bounds frame the answer. Since Γ has N eigenvalues, all in the p -element field $\mathbb{F}_p$, pigeonhole gives $m_{\max} \geq \lceil N/p \rceil$ and hence

$$r \leq N - \lceil N/p \rceil \leq N \left(1 - \frac{1}{p}\right).$$

A simple spectrum is thus impossible once $N > p$, which is the regime $N = 2^n$ reaches immediately; the correct picture is not “ $m_{\max} = O(1)$ ” but “the N eigenvalues are spread over the p available values”. If they are spread evenly, $m_{\max} \approx N/p$ and $r \approx N(1 - 1/p)$. The witness of Example 5.8 spreads them over two values and already achieves $r = N/2$.

Either way $r = \Theta(N) = \Theta(2^n)$, so the correction term $10p^{1-r}$ is of order p^{-2^n} : doubly exponentially small in the qubit count. The theorem is sharpest in exactly the regime – many qubits, expressive circuits – where the complex barren plateau is most severe.

6. Discussion

6.1. What replaces the plateau

The trainability obstruction does not vanish; it changes shape. In the complex theory it is a *measure* phenomenon: the bad set has full measure as $n \rightarrow \infty$, and no amount of care in choosing M and A escapes it. Over K , by Corollary 5.9, the bad set is the locus where the observable and the generator commute modulo p , cut out by the vanishing of a single matrix commutator over $\mathbb{F}_{p^2}$.

That locus is small but not null, and it is worth being precise about which. By Lemma 4.1 the set $\{U_B : [\widehat{M}, A] \equiv 0 \pmod{p}\}$ is a union of congruence classes, so it is open and closed in G_N and its measure is the proportion

$$\frac{\#\{g \in \mathrm{U}_N(\mathbb{F}_{p^2}) : [g^* \widetilde{M} g, \widetilde{A}] = 0\}}{|\mathrm{U}_N(\mathbb{F}_{p^2})|},$$

which is strictly positive whenever it is non-empty. It is a proper subvariety of the level-one reduction, hence of measure $O(p^{-1})$ by a Schwartz–Zippel count, but not of measure zero – a distinction with no complex analogue, and one that a finite-field ensemble makes unavoidable. The obstruction is confined to a critical subvariety rather than spread across the landscape, which matches the bifurcation structure Zúñiga-Galindo obtains for very deep classical p -adic networks [21], where the transition is between a unique network state and infinitely many rather than between a flat and a structured landscape.

There is also a design lesson, and it runs opposite to the complex one. In the complex setting expressibility hurts: the more nearly the ansatz forms a 2-design, the flatter the landscape [3]. Theorem 5.6 inverts this. Haar-randomness over G_N is what *produces* maximal gradients, because a Haar-random residue $\langle \tilde{\psi}, \Gamma \tilde{\psi} \rangle$ in $\mathbb{F}_p$ is non-zero with probability close to $1 - p^{-1}$; the danger is under-randomised circuits whose commutators sit on the critical locus. Choosing $\widehat{M}$

and A so that they visibly fail to commute modulo p is a concrete, checkable design criterion with no complex counterpart, and Example 5.8 shows the most ordinary choice already satisfies it.

6.2. Does a maximal gradient imply trainability?

Not by itself. Theorem 5.6 says the leading digit of the gradient is generically non-zero, so a step $\theta \leftarrow \theta - \eta \nabla \mathcal{L}$ with $|\eta|_p = 1$ moves the leading digit of the parameter. Convergence is governed by a separate and stringent condition. p -adic gradient descent on a quadratic loss with Hessian H contracts at rate $\|\mathbf{1} - \eta H\|$, and because the value group is discrete this is less than 1 only when every entry of $\mathbf{1} - \eta H$ lies in $p\mathbb{Z}_p$. For a *scalar* step size that says $H \equiv \eta^{-1}\mathbf{1} \pmod{p}$: the Hessian must be a scalar matrix modulo p , which is a far stronger demand than the real-analytic condition $0 < \eta < 2/\lambda_{\max}$ it replaces. With a matrix preconditioner P in place of η the requirement relaxes to invertibility of H modulo p , which is generic.

When contraction does hold, convergence is ultra-fast: the error contracts by at least p^{-1} per step, so $O(\log_p(1/\varepsilon))$ iterations suffice. Our result removes gradient concentration from the list of obstructions. It does not establish trainability, and the conditioning question is the natural next one.

6.3. Why a model with no hardware?

No p -adic quantum computer exists, and none is on any roadmap. It is worth saying what a result about such a model is for.

The first use is diagnostic, and it concerns the complex theory rather than this one. Barren plateaus are usually discussed as a phenomenon of quantum circuits, attributed to entanglement, to depth, or to the expressibility of the ansatz. The analysis above keeps the circuits, the observable, the ansatz shape and the qubit count fixed, changes only the field the amplitudes live in, and the phenomenon disappears. Whatever else barren plateaus are, they are not a consequence of unitarity or of Hilbert-space dimension alone: the load is carried by the compactness of the unitary group and by the Archimedean absolute value, two properties of $\mathbb{C}$ that the physics never singles out. This is a statement about the complex setting, and it suggests where a mitigation has to bite. Restricting the ensemble so that it is not Haar-typical in the full unitary group – the mechanism behind structured ansatz design and warm starts – acts on an ingredient the argument genuinely needs. We do not read this as a ranking of complex-side mitigations: the cost-function route of [2] attacks the same argument at a different point, and nothing here says which attack is more effective.

The second use is methodological. Replacing t -design moment combinatorics by character sums over finite unitary groups is not specific to the gradient. Lemma 4.1 reduces *any* Haar moment over G_N to counting in $U_N(\mathcal{O}/\mathfrak{m}^k)$, so second-moment questions in this model – concentration of expectation values, the distribution of overlaps, the behaviour of a p -adic kernel – become finite computations of the same shape. The design criterion the theorem yields has the same character: $[\widehat{M}, A] \not\equiv 0 \pmod{p}$ is verified by reducing two integer matrices and multiplying them, with no sampling and no asymptotics.

What the result does not do should be equally clear. It predicts nothing about any existing device, and by Remark 2.2 it cannot be checked by the empirical protocol that produces barren-plateau plots in the complex setting; the form of evidence that is available instead is set out under *No experiments* in Section 6.4. An application, if one comes, would come through the classical side of the same programme, where p -adic models are used because taxonomies, phylogenies and other hierarchical data are ultrametric to begin with [7, 8, 9, 10]. The contribution here is to that programme’s theory, not to its hardware.

6.4. Limitations

We state the boundaries of the result plainly.

- **Level one only.** Theorem 5.6 controls $\Pr[v_p(\nabla\mathcal{L}) \geq 1]$. The full distribution of $v_p(\nabla\mathcal{L})$ requires Lemma 5.3 at level $\mathfrak{m}^k$ for $k > 1$, over the finite rings $\mathcal{O}/\mathfrak{m}^k$. Lemma 4.1 supplies the reduction; the character sum has not been carried out. We expect $\Pr[v_p \geq k] = p^{-k}(1 + o(1))$, hence $\mathbb{E}[v_p(\nabla\mathcal{L})] = O(1)$, but we have not proved it.
- **The splitting hypothesis.** This is the most serious limitation. By Remark 5.5 the eigenbasis clause of Definition 5.2 is a genuine restriction and not an automatic consequence of Hermitian symmetry: over a finite field a Hermitian matrix may have eigenvalues outside the fixed field. Theorem 5.6 therefore applies to those observable–generator pairs whose commutator splits, and we do not know what happens outside that class. Example 5.8 shows the class is non-empty and contains natural pairs; it does not show the class is large.
- **Standing assumptions.** p odd and $K/\mathbb{Q}_p$ unramified. The ramified and $p = 2$ cases change the residue field and the norm group and are not covered; $p = 2$ is also the case in which the p -adic Hadamard gate does not exist [4].
- **Single parameter, two blocks.** We differentiate one parameter in the two-block ansatz (1), following [1]. Shallow-circuit and local-cost refinements in the manner of [2] are open, as is the behaviour of the full parameter vector.
- **No experiments.** By Remark 2.2 the p -adic model admits no sampling-based readout, so the usual empirical protocol does not transfer. A meaningful numerical companion would be exact arithmetic in $\mathcal{O}/\mathfrak{m}^k$ for small n and p , verifying Lemma 5.3 by enumeration; we regard this as the most useful immediate follow-up.

7. Related work

Barren plateaus were identified in [1] and refined in [2, 3]; all of that analysis is Archimedean. The p -adic quantum formalism we use is that of Aniello, Mancini and Parisi [4, 5, 6], with the tensor product of [12] and the qubit and gate constructions of Svampa and co-authors [11, 13]; the exact algebraic sense of universality goes back to Barenco et al. [22]. Haar measure on p -adic rotation groups was constructed in [23] as an inverse limit of measure spaces; our Lemma 4.1 is in the same spirit, and Proposition 3.1 explains why such constructions must target compact subgroups rather than the full unitary group, and hence why Question 1.1 is posed over G_N . On the learning side, classical p -adic neural networks go back to [7] and have been developed extensively [8, 9, 21, 10]; the quantum bridge is so far represented by the hierarchical quantum cellular network of [14]. The generalised parameter-shift rules of [24, 25] are the natural route to a p -adic gradient estimator on the finite torsion group $\mu(K)$, which our analytic $\exp_p$ formulation sidesteps but a hardware realisation would need.

8. Conclusion

Barren plateaus are a theorem about the Archimedean absolute value. Replacing $\mathbb{C}$ by a quadratic extension of $\mathbb{Q}_p$ removes them, at least on the class of circuits whose observable–generator commutator splits over $\mathbb{F}_p$. There the gradient of a p -adic variational circuit generically attains the largest absolute value the convergence radius allows, uniformly in the qubit count, and the flat locus shrinks from a full-measure set to the subvariety where the observable commutes with the generator modulo p . Whether the same holds off that class is the first question we would ask next.

Getting there required two corrections and one substitution. The ensemble had to be corrected: the full p -adic unitary group is non-compact and admits no Haar probability measure, so the

question posed over it has no answer. The scale had to be corrected: every convergent p -adic ansatz confines its one-parameter subgroups to the principal congruence subgroup, so p^{-1} , not 1, is what “maximal gradient” means. And the missing t -design combinatorics had to be replaced by counting in finite unitary groups, a substitution that appears reusable for other second-moment questions in p -adic quantum machine learning. What remains is the conditioning of the optimiser, which is now the binding constraint on trainability.

Declaration on Generative AI

The definitions, statements and proofs in this paper, and the decisions about what is correct, are the author’s own.

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